Notes on

Commutative Algebra

Lecture Notes & Problem Discussions

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Abstract

These are lecture notes for a first course in Commutative Algebra, covering rings, ideals, and modules, together with the accompanying problem-set discussions worked out alongside the lectures. The standard references for this material are [AM69], [Bly90], and [DF04].

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Chapter 1

Lecture Notes

1.1 Lecture 1

Convention. Rings are commutative with 1.

Definition 1.1. Let R, S be rings. A *morphism* φ from R to S is a map $\varphi : R \to S$ such that

(1) $\varphi(r+s) = \varphi(r) + \varphi(s)$,

(2) $\varphi(rs) = \varphi(r)\varphi(s)$,

(3) $\varphi(1) = 1$.

Convention. A subring R must contain 1.

Question. What is $R[X]$?

X is not an element of R. Consider

$$S = \{f : \mathbb{N} \to R \mid f(r) = 0 \text{ for all but finitely many } r\},$$

with the usual addition and multiplication in sequences (Cauchy product). Then S is a ring with identity

$$1_S = (1, 0, 0, \dots, 0).$$

Define $\varphi : R \to S$ by $\varphi(a) = (a, 0, \dots, 0)$. Then φ is a ring homomorphism, for sure.

Define $X = (0, 1, 0, \dots)$ and observe $X^2 = (0, 0, 1, 0, \dots)$. A polynomial P over R can then be identified as an element of S, say

$$P = (P_0, P_1, \dots, P_n, 0, \dots) = \varphi(P_0) \cdot 1_S + \varphi(P_1) \cdot X + \cdots + \varphi(P_n) \cdot X^n.$$

Definition 1.2. Let R be a ring. A non-zero, non-unit element $p \in R$ is *prime* if

$$p \mid ab \implies p \mid a \text{ or } p \mid b$$

(include the elementwise definition: for $a, b \in R$, $a \mid b$ if $\exists\, r \in R$ such that $b = r \cdot a$).

Definition 1.3. Let R be a ring. A proper ideal P of R is called a *prime ideal* if

$$ab \in P \implies a \in P \text{ or } b \in P.$$

Definition 1.4. Let R be a ring. $I \subseteq R$ is called an *ideal* if there exists a ring S and a morphism $\varphi : R \to S$ such that $I = \ker(\varphi)$.

Remark 1.5. This immediately gives the elementwise definition of ideals.

$$\mathcal{N}(R) = \{\text{all nilpotent elements of } R\} \quad \rightsquigarrow \quad \text{the } \textit{nil-radical} \text{ of } R.$$

Theorem 1.6.

$$\mathcal{N}(R) = \bigcap_{P \in \operatorname{Spec} R} P, \qquad \text{where } \operatorname{Spec} R = \text{all prime ideals of } R.$$

Proof. Let $x \in \mathcal{N}(R)$. Then for any $P \in \operatorname{Spec} R$, there exists $n_P \in \mathbb{N}$ such that

$$x^{n_P} = 0 \implies 0 \in P \implies x^{n_P} \in P \implies x \in P.$$

Now let $x \notin \mathcal{N}(R)$, and define

$$S = \{I \subseteq R \text{ ideal} \mid x^n \notin I \text{ for all } n\}.$$

$S \neq \emptyset$, as $\langle 0 \rangle \in S$, and S is a poset (ordered by inclusion). By Zorn's Lemma, let M be a maximal element of S.

Claim. M is a prime ideal.

Proof of claim. Let $a \notin M$, $b \notin M$. Now,

$$M + \langle a \rangle,\ M + \langle b \rangle \notin S \implies \exists\, m, n \in \mathbb{N} \text{ such that}$$

$$x^m \in M + \langle a \rangle \text{ and } x^n \in M + \langle b \rangle \implies x^{m+n} \in M + \langle ab \rangle.$$

If $ab \in M$, then $x^{m+n} \in M$, a contradiction. Hence $ab \notin M$, so M is prime. □

Since $x \notin M$ and $M \in \operatorname{Spec} R$, we conclude $x \notin \bigcap_{P \in \operatorname{Spec} R} P$, completing the proof. □

Question. What are the units of $R[X]$?

If R is an integral domain, then they are precisely the units of R.

Proposition 1.7. $f(X) = a_n X^n + \cdots + a_1 X + a_0$ is a unit if a_0 is a unit and the a_i's are nilpotent.

Proof. Let $P \in \operatorname{Spec} R$. We have a natural morphism $\varphi : R[X] \to (R/P)[X]$ given by

$$\varphi(b_m X^m + \cdots + b_1 X + b_0) = \overline{b_m} X^m + \cdots + \overline{b_1} X + \overline{b_0}, \qquad (\overline{b_i} = b_i \bmod P).$$

Then $\varphi(f)$ is a unit in $(R/P)[X]$, and since R/P is an integral domain, this forces

$$\overline{a_i} \equiv \overline{0} \pmod{P} \text{ for all } i \in [n], \qquad \text{and} \qquad \overline{a_0} \text{ is a unit of } R/P.$$

That is, $a_0 \notin P$ and $a_1, \ldots, a_n \in P$. As $P \in \operatorname{Spec} R$ was arbitrary,

$$a_0 \notin P \text{ and } a_i \in P \ \ \forall i \in [n], \qquad \text{for all } P \in \operatorname{Spec} R.$$

The other direction is trivial. $\square$

1.2 Lecture 2

Definition 1.8. Let R be a ring and let $(M, +)$ be an abelian group. A *scalar multiplication*

$$R \times M \longrightarrow M, \qquad (r, m) \longmapsto rm,$$

makes M into an R-*module* if the following hold for all $r, s \in R$ and $m, m_1, m_2 \in M$:

(i) $(r+s)(m) = rm + sm$,

(ii) $r(m_1 + m_2) = rm_1 + rm_2$,

(iii) $r(sm) = rsm = s(rm)$,

(iv) $1 \cdot m = m$.

Definition 1.9. If M, N are R-modules, a map $\varphi : M \to N$ is called an R-*module morphism* if

(i) $\varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2)$,

(ii) $\varphi(rm) = r\varphi(m)$.

Definition 1.10. Let R be a ring and S a nonempty set. A *free R-module on S* is a pair (F, f), where F is an R-module and $f : S \to F$ is a set-theoretic map, such that: for any R-module M and any set-theoretic map $g : S \to M$, there exists a unique R-module morphism $h : F \to M$

such that

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f"'] & M \\ F \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

commutes, i.e. $h \circ f = g$.

Proposition 1.11. Let (F, f) be a free R-module on S. Then:

(1) f is injective;

(2) $\mathrm{Im}(f)$ generates F as an R-module.

Proof of (1). Let $x, y \in S$ with $x \neq y$. **Goal:** $f(x) \neq f(y)$.

Take any R-module M with at least two elements, and define $g : S \to M$ such that $g(x) \neq g(y)$. By definition, there exists a (unique) morphism $h : F \to M$ such that $h \circ f = g$. Then

$$(h \circ f)(x) = g(x) \neq g(y) = (h \circ f)(y) \implies f(x) \neq f(y). \qquad \square$$

Definition 1.12. Let M be an R-module and $\emptyset \neq A \subseteq M$. The *submodule of M generated by A* is, equivalently:

(1) the smallest submodule of M containing A;

(2) the intersection of all submodules of M containing A;

(3) the set of all finite R-linear combinations of elements of A.

Proof of (2). Let A be the submodule of F generated by $\mathrm{Im}(f)$. Since $f(s) \in \mathrm{Im}(f) \subseteq A$ for every $s \in S$, f factors as $f = i_A \circ f^*$ for a set-theoretic map $f^* : S \to A$ and the inclusion $i_A : A \hookrightarrow F$:

$$\begin{tikzcd} S \arrow[r, "f^*"] \arrow[dr, "f"'] & A \arrow[d, hook, "i_A"] \\ & F \end{tikzcd}$$

As (F, f) is free on S and A is an R-module, there is a unique morphism $h : F \to A$ such that $h \circ f = f^*$:

$$\begin{tikzcd} S \arrow[r, "f^*"] \arrow[d, "f"'] & A \\ F \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

It is enough to show i_A is onto. We compute

$$(i_A \circ h) \circ f = i_A \circ (h \circ f) = i_A \circ f^* = f.$$

So $i_A \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness in the definition of freeness (applied with $M = F, g = f$), $i_A \circ h = \mathrm{id}_F$. Hence i_A is onto, i.e. $A = F$, so $\mathrm{Im}(f)$ generates F. $\square$

Theorem 1.13: Uniqueness of free modules. Let (F, f) be a free R-module on a set S. Then (F', f') is also a free R-module on S if and only if there exists a unique R-module isomorphism $j : F \to F'$ such that $j \circ f = f'$.

Proof. Suppose (F', f') is also free on S. Since (F, f) is free (applied to $M = F'$, $g = f'$), there is a morphism $h : F \to F'$ with $h \circ f = f'$:

$$\begin{tikzcd} S \arrow[r, "f'"] \arrow[d, "f"'] & F' \\ F \arrow[ur, "h"'] & \end{tikzcd}$$

Likewise, since (F', f') is free (applied to $M = F$, $g = f$), there is a morphism $h' : F' \to F$ with $h' \circ f' = f$:

$$\begin{tikzcd} S \arrow[r, "f"] \arrow[d, "f'"'] & F \\ F' \arrow[ur, "h'"'] & \end{tikzcd}$$

Then $(h' \circ h) \circ f = h' \circ (h \circ f) = h' \circ f' = f$. So $h' \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness (with $M = F, g = f$), $h' \circ h = \mathrm{id}_F$. Similarly $h \circ h' = \mathrm{id}_{F'}$. So $j := h$ is an isomorphism with $j \circ f = f'$.

Converse. Suppose there is an isomorphism $j : F \to F'$ with $j \circ f = f'$. To show (F', f') is free on S, take any R-module M and set-theoretic map $g : S \to M$; we must produce a unique $h' : F' \to M$ with $h' \circ f' = g$:

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f'"'] & M \\ F' \arrow[ur, dashed, "h'"'] & \end{tikzcd}$$

Since (F, f) is free, there is a unique $h : F \to M$ with $h \circ f = g$. From $j \circ f = f'$ we get $j^{-1} \circ f' = f$, so

$$(h \circ j^{-1}) \circ f' = h \circ (j^{-1} \circ f') = h \circ f = g,$$

i.e. $h' := h \circ j^{-1}$ works:

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f'"'] \arrow[dr, "f"'] & M \\ F' \arrow[r, "j^{-1}"'] & F \arrow[u, "h"'] \end{tikzcd}$$

Uniqueness. Suppose $t : F' \to M$ also satisfies $t \circ f' = g$. Then $t \circ j : F \to M$ satisfies $(t \circ$

$j) \circ f = t \circ (j \circ f) = t \circ f' = g$, so by uniqueness of h (from freeness of (F, f)), $t \circ j = h$, hence $t = h \circ j^{-1} = h'$. □

Theorem 1.14: Existence of free R-modules. For any ring R and nonempty set S, a free R-module on S exists.

Proof. Define
$$F = \{\theta : S \to R \mid \theta(s) = 0 \text{ for almost all } s \in S\}.$$

For $\theta_1, \theta_2 \in F$ and $\lambda \in R$, define

$$(\theta_1 + \theta_2)(s) = \theta_1(s) + \theta_2(s), \qquad (\lambda\theta)(s) = \lambda\,\theta(s).$$

One checks that F is an R-module. Define $f : S \to F$ by $f(s) = \delta_s$, where $\delta_s(t) = 1$ if $t = s$ and 0 otherwise.

Existence of h. Let M be an R-module and $g : S \to M$ a set-theoretic map. Define $h : F \to M$ by

$$h(\theta) = \sum_{s \in S} \theta(s)\, g(s),$$

a finite sum for every $\theta \in F$. For any $t \in S$,

$$(h \circ f)(t) = h(f(t)) = h(\delta_t) = \delta_t(t)\, g(t) = g(t),$$

so $h \circ f = g$:

$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "f"'] & M \\ F \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

Uniqueness of h. For $\theta \in F$ and $t \in S$,

$$\theta(t) = \theta(t) \cdot 1_R = \sum_{s \in S} \theta(s)\, \delta_s(t) = \left(\sum_{s \in S} \theta(s)\, \delta_s \right)(t) \qquad \forall\, t \in S,$$

so $\theta = \sum_{s \in S} \theta(s)\, \delta_s$. Now let $h' : F \to M$ be any R-module morphism with $h' \circ f = g$. Then for $\theta \in F$,

$$h'(\theta) = h' \left(\sum_{s \in S} \theta(s)\, \delta_s \right) = \sum_{s \in S} \theta(s)\, h'(\delta_s) = \sum_{s \in S} \theta(s)\, h'(f(s)) = \sum_{s \in S} \theta(s)\, g(s) = h(\theta).$$

Hence $h = h'$. □

1.3 Lecture 3

Theorem 1.15. Let R be a ring such that every prime ideal is principal. Then every ideal is principal.

Proof. Let $S = \{\text{all non-principal ideals of } R\}$. For the sake of contradiction (FTSOC), suppose $S \neq \emptyset$. By Zorn's Lemma, let I be a maximal element of S.

Since I is not principal, I is not prime by hypothesis, so there exist $a, b \in R$ with $a \notin I$, $b \notin I$, but $ab \in I$.

Let $J = \langle a \rangle + I \supsetneq I$; since I is maximal among non-principal ideals, $J \notin S$, so J is principal, say $J = \langle c \rangle$.

Also let $K = \{r \in R \mid ra \in I\} \supsetneq I$ (since $b \in K \setminus I$); again $K \notin S$, so $K = \langle d \rangle$ for some d.

$I \subseteq \langle cd \rangle$. Let $x \in I$. Since $I \subseteq J = \langle c \rangle$, there is $\lambda \in R$ with $\lambda c = x$. Now $\lambda a \in I$, so $\lambda \in K = \langle d \rangle$, and hence $x \in \langle cd \rangle$. Thus $I \subseteq \langle cd \rangle$.

$\langle cd \rangle \subseteq I$. There exist $u \in R$, $v \in I$ such that $c = ua + v$ (since c generates $J = \langle a \rangle + I$). Also $da \in I$ (since d generates K). Then

$$cd = (ua + v)d = u(ad) + vd \in I,$$

since $ad = da \in I$ and $v \in I$. Hence $\langle cd \rangle \subseteq I$.

Together, $I = \langle cd \rangle$, contradicting $I \in S$. So $S = \emptyset$, i.e. every ideal of R is principal. $\square$

Theorem 1.16. Let R be a ring such that every prime ideal is finitely generated. Then every ideal of R is finitely generated.

Proof. Let $A = \{I \text{ ideal of } R \mid I \text{ is not finitely generated}\}$. By Zorn's Lemma, let M be a maximal element of A.

Claim: M is a maximal ideal. Suppose not, so there is an ideal N with $M \subsetneq N \subsetneq R$. Since N is maximal (hence prime), it is finitely generated by hypothesis: there exist $a_1, \ldots, a_n \in R$ such that $N = \langle a_1, \ldots, a_n \rangle$.

Since $M \subsetneq N$, some $a_i \notin M$. Then $M + \langle a_i \rangle \supsetneq M$, so by maximality of M in A, $M + \langle a_i \rangle \notin A$; hence there exist $b_1, \ldots, b_k \in R$ such that

$$M + \langle a_i \rangle = \langle b_1, \ldots, b_k \rangle.$$

$\square$

Lemma 1.17. Let I, J_1, J_2 be ideals of R such that $I \subseteq J_1 \cup J_2$. Then either $I \subseteq J_1$ or $I \subseteq J_2$.

Proof. FTSOC, suppose $I \setminus J_1 \neq \emptyset$ and $I \setminus J_2 \neq \emptyset$. Let $x \in I \setminus J_1$ and $y \in I \setminus J_2$; since $I \subseteq J_1 \cup J_2$, this forces $x \in J_2$ and $y \in J_1$. Also $x + y \in I \subseteq J_1 \cup J_2$, so $x + y \in J_1$ or $x + y \in J_2$; WLOG $x + y \in J_1$. Since $y \in J_1$, $x = (x + y) - y \in J_1$, contradicting $x \notin J_1$. $\square$

Theorem 1.18: Prime Avoidance. Let $I, J_1, \ldots, J_n$ be ideals of R such that

(1) $I \subseteq J_1 \cup \cdots \cup J_n$,

(2) $J_3, \ldots, J_n \in \operatorname{Spec} R$.

Then $I \subseteq J_i$ for some i.

Proof. We induct on n. The case $n = 2$ needs no primality hypothesis and is the lemma above; assume $n \geq 3$ and the result for any collection of $n - 1$ such ideals.

We prove the contrapositive: if $I \not\subseteq J_i$ for every i, then $I \not\subseteq \bigcup_{i=1}^n J_i$.

Fix i. Apply the induction hypothesis to I together with the $n-1$ ideals $\{J_j\}_{j \neq i}$ (which still has all but at most two members prime): if $I \subseteq \bigcup_{j \neq i} J_j$, the induction hypothesis would give $I \subseteq J_j$ for some $j \neq i$, contradicting our assumption. So $I \not\subseteq \bigcup_{j \neq i} J_j$, and we may choose

$$z_i \in I \setminus \bigcup_{j \neq i} J_j.$$

Since $I \subseteq \bigcup_{i=1}^n J_i$ by hypothesis (1), and $z_i \notin J_j$ for every $j \neq i$, we must have $z_i \in J_i$, for every i.

Set

$$z := z_1 z_2 \cdots z_{n-1} + z_n \in I$$

(a sum of an element of I and a product of elements of I, hence itself in I). We show $z \notin J_i$ for every i, which gives $I \not\subseteq \bigcup_{i=1}^n J_i$, as desired.

- *If $i \in \{1, \ldots, n-1\}$:* since $z_i \in J_i$ is one of the factors of $z_1 \cdots z_{n-1}$, this product lies in J_i. If $z \in J_i$, then $z_n = z - z_1 \cdots z_{n-1} \in J_i$, contradicting $z_n \notin J_i$ (as $i \neq n$).

- *If $i = n$:* if $z \in J_n$, then since $z_n \in J_n$ as well, $z_1 \cdots z_{n-1} = z - z_n \in J_n$. As J_n is prime, some factor $z_k \in J_n$ for a $k \in \{1, \ldots, n-1\}$, contradicting $z_k \notin J_n$ (as $k \neq n$).

This completes the induction. □

Proposition 1.19. An element of $\mathbb{Q}(i)$ lies in $\mathbb{Z}[i]$ if and only if it is the root of a monic polynomial over $\mathbb{Z}$.

Proof. Let $a + bi \in \mathbb{Q}(i)$. Then $a + bi$ is a root of

$$P(X) = X^2 - 2aX + (a^2 + b^2) \in \mathbb{Q}[X].$$

Claim: $2a \in \mathbb{Z}$ and $a^2 + b^2 \in \mathbb{Z}$ together imply $a, b \in \mathbb{Z}$.

Since $2a \in \mathbb{Z}$, $(2a)^2 \in \mathbb{Z}$. Since $a^2 + b^2 \in \mathbb{Z}$, $4(a^2 + b^2) = (2a)^2 + (2b)^2 \in \mathbb{Z}$, so $(2b)^2 = 4(a^2 + b^2) - (2a)^2 \in \mathbb{Z}$. As $2b \in \mathbb{Q}$ has integer square, $2b \in \mathbb{Z}$.

Now $(2a)^2 + (2b)^2 = 4(a^2 + b^2) \equiv 0 \pmod 4$. Since the square of any integer is $\equiv 0 \pmod 4$ if it is even, and $\equiv 1 \pmod 4$ if it is odd, the only way for the sum to be $\equiv 0 \pmod 4$ is

$$(2a)^2 \equiv (2b)^2 \equiv 0 \pmod 4,$$

forcing $2a, 2b$ to both be even, i.e. $a, b \in \mathbb{Z}$.

Since $a + bi \in \mathbb{Z}[i]$ exactly when $a, b \in \mathbb{Z}$, and $a + bi$ is a root of a monic polynomial over $\mathbb{Z}$ exactly when $2a, a^2 + b^2 \in \mathbb{Z}$ (the coefficients of $P(X)$), the claim gives the equivalence. □

Definition 1.20. Let $R \hookrightarrow S$ be rings. An element $\alpha \in S$ is called *integral* over R if α is a root of a monic polynomial

$$X^n + a_{n-1}X^{n-1} + \cdots + a_0$$

over R (i.e. with $a_0, \ldots, a_{n-1} \in R$).

1.4 Lecture 4

Definition 1.21. Let R be a subring of S, and let $U \subseteq S$ be a subset. Define

$$R[U] := \text{the smallest subring of } S \text{ containing } R \text{ and } U.$$

Remark 1.22. For $U, V \subseteq S$,

$$R[U][V] = R[U \cup V] = R[V][U].$$

In particular, if $V = \{u_1, \ldots, u_n\}$, then

$$R[u_1, \ldots, u_n] = R[u_1][u_2] \cdots [u_n].$$

Proposition 1.23.

$$R[u] = \{a_0 + a_1 u + \cdots + a_n u^n \mid a_i \in R,\ n \in \mathbb{N}_{\geq 0}\}.$$

Proof. Let $T = \{a_0 + a_1 u + \cdots + a_n u^n \mid a_i \in R,\ n \in \mathbb{N}_{\geq 0}\}$.

T **is a subring of** S**.** It contains 1 (take $n = 0$, $a_0 = 1$). It is closed under addition: given two such expressions, pad the shorter one with zero coefficients to a common degree and add coefficientwise, which stays in R since R is a ring. It is closed under multiplication: expanding a product of two such expressions and collecting powers of u, each resulting coefficient is a finite sum of products of elements of R, hence lies in R.

$R[u] \subseteq T$**.** Since $R \subseteq T$ (take $n = 0$) and $u \in T$ (take $n = 1, a_0 = 0, a_1 = 1$), and T is a subring of S containing R and u, while $R[u]$ is by definition the *smallest* such subring, we get $R[u] \subseteq T$.

$T \subseteq R[u]$**.** Since $R[u]$ is a subring containing R and u, it is closed under addition and multiplication. So for any $a_0, \ldots, a_n \in R \subseteq R[u]$, each $a_i u^i \in R[u]$, and hence $a_0 + a_1 u + \cdots + a_n u^n \in R[u]$.

Combining both containments, $R[u] = T$. $\square$

Example 1.24. The map $\mathbb{Z}[X] \to \mathbb{Z}[i]$, $X \mapsto i$, is a surjective ring morphism with kernel $\langle X^2 + 1 \rangle$.

Proposition 1.25. Let R, S be rings and $\varphi : R \to S$ be a morphism. Let $\alpha \in S$ be an element. Then φ can be uniquely extended to a morphism $\widetilde{\varphi} : R[X] \to S$ such that $\widetilde{\varphi}|_R = \varphi$ and $\widetilde{\varphi}(X) = \alpha$.

Proof sketch. Define $\widetilde{\varphi} : R[X] \to S$ by

$$\widetilde{\varphi}(a_0 + a_1 X + \cdots + a_n X^n) = \varphi(a_0) + \varphi(a_1)\alpha + \cdots + \varphi(a_n)\alpha^n,$$

and $\widetilde{\varphi}(A \cdot B) = \widetilde{\varphi}(A) \cdot \widetilde{\varphi}(B)$. $\square$

Let R be a subring of S, and let $u \in S$. Then there is a unique morphism $\varphi : R[X] \to S$ such that $\varphi|_R = \mathrm{id}_R$ and $\varphi(X) = u$ (apply the above with $\varphi = \mathrm{id}_R$, $\alpha = u$). Then $\mathrm{Im}(\varphi) = R[u]$; letting $I = \ker \varphi$,

$$R[X]/I \cong R[u].$$

Notation. $I \cap R = \langle 0 \rangle$.

Conversely, let I be an ideal of $R[X]$ such that $I \cap R = \langle 0 \rangle$. We have the natural projection map

$$R[X] \longrightarrow R[X]/I.$$

Since $I \cap R = 0$, $R \hookrightarrow R[X]/I$, so we can identify $a \in R$ with its image $a + I$.

Question. What is the image of $R[X]$ in $R[X]/I$? (It will be determined by $\mathrm{Im}(R)$ and that of X.)

Under the projection, $X \mapsto X + I$. Writing $u = X + I$,

$$R[X]/I \cong R[u].$$

Definition 1.26. Let $R \hookrightarrow S$. S is said to be an *integral extension* of R if each $\alpha \in S$ is integral over R.

Notation. (Recall.) Let R be a commutative ring with 1, and let $A \in M_n(R)$. Then

$$A \cdot \mathrm{adj}(A) = \mathrm{adj}(A) \cdot A = \det(A) \cdot I_n.$$

Theorem 1.27. Let $R \hookrightarrow S$ and $b \in S$. Then

$$b \text{ is integral over } R \iff R[b] \text{ is a finitely generated } R\text{-module.}$$

Proof. ($\Rightarrow$) Assume b is integral over R; then there is a monic polynomial $f(X) \in R[X]$ such that

$f(b) = 0$. Let $\deg f = n \geq 1$. Take any $g(b) \in R[b]$. By the division algorithm,

$$g(X) = q(X)f(X) + r(X), \qquad \deg r < n,$$

so $g(b) = r(b)$. Hence $R[b]$ is generated (as an R-module) by $\{1, b, \ldots, b^{n-1}\}$.

($\Leftarrow$) Conversely, let $R[b]$ be a finitely generated R-module, say $R[b] = \langle w_1, \ldots, w_n \rangle$. Then for each $i \in [n]$,

$$bw_i = a_{i1}w_1 + \cdots + a_{in}w_n, \qquad a_{ij} \in R,$$

i.e. $(bI - A)\vec{w} = 0$, where $A = (a_{ij})$. Multiplying by $\operatorname{adj}(bI - A)$,

$$\det(bI - A)\, w_i = 0 \qquad \forall\, i \in [n].$$

Since $1 \in R[b] = \langle w_1, \ldots, w_n \rangle$, we get $\det(bI - A) = 0$, which is a monic equation in b (of degree n). Hence b is integral over R. $\square$

Theorem 1.28. Let $R \hookrightarrow S$, and let $b_1, b_2, \ldots, b_n \in S$. Then each b_i is integral over R if and only if $R[b_1, \ldots, b_n]$ is a finitely generated R-module.

Proof. Induct on n. The case $n = 1$ is the previous theorem.

For $n > 1$: assume $b_1, \ldots, b_n$ are integral over R. In particular b_n is integral over R, hence also integral over $R[b_1, \ldots, b_{n-1}]$; by the previous theorem (applied over the base ring $R[b_1, \ldots, b_{n-1}]$), $R[b_1, \ldots, b_{n-1}][b_n]$ is a finitely generated $R[b_1, \ldots, b_{n-1}]$-module, which (combined with the induction hypothesis that $R[b_1, \ldots, b_{n-1}]$ is a finitely generated R-module) makes $R[b_1, \ldots, b_n]$ a finitely generated R-module.

The converse is the same as the $n = 1$ case. $\square$

Exercise 1.29. Let $R \hookrightarrow S$, and let $a, b \in S$ be integral over R. Then $a \pm b$ and ab are integral over R.

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Corollary 1.30. Let $R \hookrightarrow S$ and $b_1, \ldots, b_n \in S$ be integral over R (equivalently, $R[b_1, \ldots, b_n]$ is a finitely generated R-module). Then every $b \in R[b_1, \ldots, b_n]$ is integral over R.

Proof. Since $b \in R[b_1, \ldots, b_n]$,

$$R[b_1, \ldots, b_n] = R[b_1, \ldots, b_n, b],$$

which is a finitely generated R-module. By the previous theorem, b is integral over R. $\square$

Proposition 1.31. Let $A \subseteq B \subseteq C$ be rings such that B is integral over A and C is integral over B. Then C is integral over A.

Proof. Let $c \in C$. Then c is integral over B:

$$c^n + b_{n-1}c^{n-1} + \cdots + b_0 = 0, \qquad b_i \in B.$$

Let $R := A[b_0, \ldots, b_{n-1}]$. Then c is integral over R, so $R[c]$ is a finitely generated R-module.

Since $b_0, \ldots, b_{n-1}$ are integral over A, R is a finitely generated A-module. Hence $R[c]$ is a finitely generated A-module, and since $c \in R[c]$, c is integral over A. □

Definition 1.32. Define $\overline{R}^S := \{r \in S \mid r \text{ is integral over } R\}$. Note that $\overline{R}^S$ is a subring of S. This $\overline{R}^S$ is called the *integral closure* of R in S.

Definition 1.33. R is said to be *integrally closed in* S if $\overline{R}^S = R$.

$$R \hookrightarrow \overline{R}^S \hookrightarrow S.$$

Observation 1.34. $\overline{R}^S$ is integrally closed in S.

Definition 1.35. Let R be an integral domain with quotient field $F = \mathbb{Q}(R)$. We say that R is *integrally closed* if R is integrally closed in F.

Proposition 1.36. If R is a UFD, then R is integrally closed.

Proof. Let $F = \mathbb{Q}(R)$. Say $r/s \in F$ with $r, s \in R$, $s \neq 0$, is integral over R, and suppose r, s have no common prime factors. Then

$$\left(\frac{r}{s}\right)^n + a_{n-1}\left(\frac{r}{s}\right)^{n-1} + \cdots + a_1\left(\frac{r}{s}\right) + a_0 = 0, \qquad a_i \in R,$$

so

$$r^n + a_{n-1}r^{n-1}s + \cdots + a_1 rs^{n-1} + a_0 s^n = 0.$$

Since s divides every term but the first, $s \mid r^n$. FTSOC, if s is not a unit, it has a prime factor p; then $p \mid r^n$, and since p is prime, $p \mid r$ – contradicting that r, s have no common prime factors. Hence s is a unit in R, so $r/s \in R$. □

Example 1.37. $\mathbb{Z}[\sqrt{-5}]$ is integrally closed but not a UFD. (Exercise.)

Example 1.38. $\mathbb{Z}[\sqrt{5}]$ is an integral domain which is not integrally closed.

Proof. $\mathbb{Q}(\mathbb{Z}[\sqrt{5}]) = \mathbb{Q}(\sqrt{5})$, and
$$\frac{1+\sqrt{5}}{2} \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Z}[\sqrt{5}]$$
is a root of the monic polynomial $X^2 - X - 1 \in \mathbb{Z}[X]$, hence integral over $\mathbb{Z}[\sqrt{5}]$, yet not itself an element of $\mathbb{Z}[\sqrt{5}]$. $\square$

Extensions and Contractions

Let $\varphi : R \to S$ be a ring morphism. Then φ factors as
$$R \xrightarrow{\varphi} \varphi(R) \xhookrightarrow{i} S \qquad \text{with} \qquad R/\operatorname{Ker}\varphi \cong \varphi(R).$$

(1) If $I \subseteq R$ is an ideal, $\varphi(I)$ need *not* be an ideal of S.

(2) If $J \subseteq S$ is an ideal, $\varphi^{-1}(J)$ *is* an ideal of R.

(3) If $Q \in \operatorname{Spec} S$, then $\varphi^{-1}(Q) \in \operatorname{Spec} R$.

Question. Suppose $P \in \operatorname{Spec} R$. Can we say that $\langle \varphi(P) \rangle$ is prime in S?

Consider $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$: $\langle 2 \rangle \mapsto \langle 1+i \rangle \langle 1-i \rangle$. If $p \equiv 1 \pmod 4$ is prime, then $\langle p \rangle$ is *not* prime in $\mathbb{Z}[i]$ (so the answer to the question above is no, in general).

Exercise 1.39. If $p \equiv 3 \pmod 4$, then $\langle p \rangle$ is a prime ideal in $\mathbb{Z}[i]$.

Definition 1.40. Let $\varphi : R \to S$ be a ring morphism.

- For I an ideal of R, the *extension* I^e is the ideal of S generated by $\varphi(I)$.
- For J an ideal of S, the *contraction* $J^c := \varphi^{-1}(J)$, an ideal of R.

Exercise 1.41.
$$I \subseteq I^{ec}, \qquad J \supseteq J^{ce}, \qquad I^e = I^{ece}, \qquad J^c = J^{cec}.$$

Rings of Fractions

Definition 1.42. Let R be a ring. A subset $S \subseteq R$ is called *multiplicative* if

(1) $1 \in S$,

(2) $a, b \in S \implies ab \in S$.

Construction 1.43. On $R \times S$, define the relation $\sim$ by

$$(a,s) \sim (b,t) \iff \exists\, u \in S \text{ such that } u(at - bs) = 0.$$

$\sim$ is transitive. Suppose $(a,s) \sim (b,t)$ and $(b,t) \sim (c,u)$; so $(at - bs)v = 0$ for some $v \in S$, and $(bu - ct)w = 0$ for some $w \in S$. Then

$$atv - bsv = 0 \rightsquigarrow uw{\cdot}atv - bsuw = 0, \qquad buw - ctw = 0 \rightsquigarrow buwsv - ctwsv = 0.$$

Adding,

$$(au - cs)\, vtw = 0, \qquad vtw \in S,$$

so $(a,s) \sim (c,u)$. $\square$

Notation. The equivalence class of (a,s) is denoted $\dfrac{a}{s}$.

Let

$$S^{-1}R := \{\text{all equivalence classes}\} = \left\{ \frac{a}{s} \;\middle|\; a \in R,\ s \in S \setminus \{0\} \right\}.$$

Define

$$\frac{a}{s} + \frac{b}{t} := \frac{at + bs}{st} \quad \text{(well-defined)}, \qquad \frac{a}{s} \cdot \frac{b}{t} := \frac{ab}{st} \quad \text{(well-defined)}.$$

$S^{-1}R$ is a commutative ring with 1. We have a canonical ring morphism

$$\varphi : R \to S^{-1}R, \qquad r \mapsto \frac{r}{1}.$$

Remark 1.44. φ is not injective in general.

Proposition 1.45. Let $g : R \to A$ be a ring morphism such that $g(S) \subseteq A^\times$ (the units of A). Then there is a unique morphism $h : S^{-1}R \to A$ such that

$$\begin{tikzcd} R \arrow[r, "g"] \arrow[d, "\varphi"'] & A \\ S^{-1}R \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

commutes, i.e. $h \circ \varphi = g$.

Proof. **Uniqueness.** Suppose such an h exists. Then

$$h\left(\frac{r}{1}\right) = h(\varphi(r)) = g(r), \qquad h\left(\frac{1}{s}\right) = h\left(\left(\frac{s}{1}\right)^{-1}\right) = h\left(\frac{s}{1}\right)^{-1} = g(s)^{-1},$$

so

$$h\left(\frac{r}{s}\right) = h\left(\frac{r}{1}\right) h\left(\frac{1}{s}\right) = g(r)\, g(s)^{-1},$$

i.e. h is uniquely determined by g.

Existence. Define $h : S^{-1}R \to A$ by $h(r/s) = g(r)g(s)^{-1}$.

Well-definedness. Suppose $r/s = r'/s'$; then there is $t \in S$ with $t(rs' - r's) = 0$, so

$$g(t)\big(g(r)g(s') - g(r')g(s)\big) = 0.$$

Since $g(t) \in A^\times$, $g(r)g(s') = g(r')g(s)$, i.e. $g(r)g(s)^{-1} = g(r')g(s')^{-1}$. □

Example 1.46. Let $P \in \operatorname{Spec} R$ and $S = R \setminus P$. Then S is multiplicative; write $S^{-1}R =: R_P$. Let

$$PR_P := \left\{ \frac{p}{s} \,\middle|\, p \in P,\ s \in R \setminus P \right\},$$

an ideal of R_P. If $b/t \notin PR_P$, then $b \notin P$, so $b \in R \setminus P$, so b/t is a unit in R_P. Hence PR_P is the unique maximal ideal of R_P, so R_P is local.

Example 1.47. Let $f \in R$ and $S = \{1, f, f^2, \dots\}$. Then $R_f := S^{-1}R$.

Construction 1.48. Let R be a ring, S a multiplicative subset of R, and M an R-module. On $M \times S$, define $\sim$ by

$$(m,s) \sim (m',s') \iff \exists\, t \in S \text{ such that } t(ms' - m's) = 0.$$

Then $S^{-1}M := (M \times S)/{\sim}$ is an $S^{-1}R$-module.

Let M, N be R-modules and $\varphi : M \to N$ an R-module morphism. Define $S^{-1}\varphi : S^{-1}M \to S^{-1}N$ by

$$\frac{m}{s} \longmapsto \frac{\varphi(m)}{s}.$$

$S^{-1}\varphi$ is a well-defined $S^{-1}R$-module morphism.

Proposition 1.49. $S^{-1}(-)$ preserves exactness: if

$$M' \xrightarrow{\varphi} M \xrightarrow{\psi} M''$$

is exact at M, then

$$S^{-1}M' \xrightarrow{S^{-1}\varphi} S^{-1}M \xrightarrow{S^{-1}\psi} S^{-1}M''$$

is exact at $S^{-1}M$.

Proof. We have $\psi \circ \varphi = 0$, so $(S^{-1}\psi) \circ (S^{-1}\varphi) = S^{-1}(\psi \circ \varphi) = 0$. Hence $\operatorname{Im}(S^{-1}\varphi) \subseteq \operatorname{Ker}(S^{-1}\psi)$.

Conversely, let $m/s \in \operatorname{Ker}(S^{-1}\psi)$. Then $\psi(m)/s = 0$, so there is $t \in S$ with $t\,\psi(m) = 0$, i.e.

$\psi(tm) = 0$, i.e. $tm \in \operatorname{Ker}(\psi) = \operatorname{Im}(\varphi)$. So there is $m' \in M'$ with $\varphi(m') = tm$. Then

$$\frac{m}{s} = \frac{mt}{st} = \frac{\varphi(m')}{st} = (S^{-1}\varphi)\left(\frac{m'}{st}\right) \in \operatorname{Im}(S^{-1}\varphi).$$

Hence $\operatorname{Ker}(S^{-1}\psi) \subseteq \operatorname{Im}(S^{-1}\varphi)$, giving equality. ¡¡¡¡¡¡ HEAD:chapters/lecture-notes.tex ======= □

1.6 Lecture 6

Theorem 1.50. Let M be an R-module generated by n elements, and let

$$\varphi \in \operatorname{Hom}(M, M) := \{f : M \to M \mid f \text{ is an } R\text{-module morphism}\}.$$

Let I be an ideal of R such that $\varphi(M) \subseteq IM$. Then there is a relation of the form

$$\varphi^n + a_{n-1}\varphi^{n-1} + \cdots + a_0 = 0, \qquad a_i \in I^i.$$

Proof. Let $M = Rw_1 + \cdots + Rw_n$, so $IM = Iw_1 + \cdots + Iw_n$; in particular $\varphi(M) \subseteq IM$. Write

$$\varphi(w_i) = \sum_{j=1}^{n} a_{ij} w_j, \qquad a_{ij} \in I \text{ for all } i,$$

i.e.

$$\sum_{j=1}^{n} (\varphi\,\delta_{ij} - a_{ij}) w_j = 0.$$

The coefficient matrix has (i, j)-entry $\varphi\,\delta_{ij} - a_{ij}$, with entries in the commutative subring of $\operatorname{End}(M)$ generated by R (acting by scalars) and φ. Let $d = \det(\text{coefficient matrix})$. Multiplying by the adjugate matrix,

$$d \cdot w_k = 0 \qquad \forall\, k \in [n],$$

so $d \cdot M = 0$, i.e. d is the zero endomorphism of M. Expanding the determinant gives exactly a relation of the stated form. □

Theorem 1.51: Nakayama's Lemma. Let M be a finitely generated R-module and I an ideal of R. If $M = IM$, then there exists $a \in R$ such that $aM = 0$ and $a \equiv 1 \pmod{I}$. Furthermore, if I is the Jacobson radical of R, then $M = 0$.

Proof. Take $\varphi = \mathrm{id}_M$. By the previous theorem, we have

$$1 + a_{n-1} + \cdots + a_0 = 0 \qquad (\text{as endomorphisms of } M).$$

Define $a := 1 + a_{n-1} + \cdots + a_0$. Then $aM = 0$ and $a \equiv 1 \pmod{I}$.

If $I \subseteq J(R)$ (the Jacobson radical), then $a - 1 \in J(R)$, so a is a unit; hence $aM = 0$ forces $M =$

$a^{-1}(aM) = 0$. □

Theorem 1.52. Let M be a finitely generated R-module and I an ideal of R such that $M = IM$. Then there exists $i \in I$ such that $(1+i)M = 0$.

Proof (Ojanguren). Induct on the minimal number of generators $\nu(M) = n$ of M.

If $\nu(M) = 0$, then $M = 0$ and we are done.

Let $M = Rm_1 + \cdots + Rm_n$, and set $M' := M/Rm_n$. Then

$$IM' = \frac{IM + Rm_n}{Rm_n} = \frac{M + Rm_n}{Rm_n} = \frac{M}{Rm_n} = M',$$

using $IM = M$. Since $\nu(M') < n$, by induction there is $x \in I$ such that $(1+x)M' = 0$, i.e.

$$(1+x)\frac{M}{Rm_n} = 0 \qquad \Longrightarrow \qquad (1+x)M \subseteq Rm_n.$$

Then

$$(1+x)M = (1+x)IM = I(1+x)M \subseteq I \cdot Rm_n = Im_n,$$

using $(1+x)M \subseteq Rm_n$ in the last containment. In particular $(1+x)m_n \in Im_n$, so $(1+x)m_n = ym_n$ for some $y \in I$, giving

$$(1+x-y)m_n = 0.$$

Now for any $\mu \in M$, $(1+x)\mu = rm_n$ for some $r \in R$ (since $(1+x)M \subseteq Rm_n$), so

$$(1+x-y)(1+x)\mu = r(1+x-y)m_n = 0.$$

Hence $(1+x)(1+x-y)M = 0$. Expanding, $(1+x)(1+x-y) = 1+i$ where $i := x + (x-y) + x(x-y) \in I$. So $(1+i)M = 0$. □

Definition 1.53. Let M, N be R-modules. The *tensor product* of M, N is a pair (T, θ) such that T is an R-module, $\theta : M \times N \to T$ is bilinear, and for every bilinear map $f : M \times N \to P$, there is a unique linear map $h : T \to P$ such that

$$\begin{tikzcd} M \times N \arrow[r, "f"] \arrow[d, "\theta"'] & P \\ T \arrow[ur, dashed, "h"'] & \end{tikzcd}$$

commutes, i.e. $h \circ \theta = f$.

Recall the canonical morphism $R \to S^{-1}R$, $r \mapsto r/1$, and let I be an ideal of R. An element of the extended ideal I^e is a finite sum

$$\frac{r_1}{s_1} \cdot \frac{x_1}{1} + \cdots + \frac{r_n}{s_n} \cdot \frac{x_n}{1} = \frac{s_1' r_1 x_1 + \cdots + s_n' r_n x_n}{s_1 \cdots s_n},$$

where $r_i \in R$, $s_i \in S$, $x_i \in I$ (writing s_i' for the product of all s_j with $j \neq i$).

Proposition 1.54.

(1) Every ideal of $S^{-1}R$ is an extended ideal.

(2) The prime ideals of $S^{-1}R$ are in one-to-one correspondence, $P \leftrightarrow S^{-1}P$, with the prime ideals P of R such that $S \cap P = \emptyset$.

Proof. See [AM69]. □

Local Properties of Modules

Proposition 1.55. Let M be an R-module. Then the following are equivalent:

(1) $M = 0$;

(2) $M_P = 0$ for all $P \in \operatorname{Spec}(R)$;

(3) $M_{\mathfrak{m}} = 0$ for all $\mathfrak{m} \in \operatorname{Max}(R)$.

(Here for $P \in \operatorname{Spec}(R)$, writing $S = R \setminus P$, we set $M_P := S^{-1}M$.)

Proof. (1) $\Rightarrow$ (2) $\Rightarrow$ (3) trivially.

(3) $\Rightarrow$ (1): Let $M \neq 0$; then there is $x \in M$ with $x \neq 0$. Let $I = \operatorname{Ann}(x) = \{r \in R \mid rx = 0\}$, a proper ideal of R. Let $\mathfrak{m}$ be a maximal ideal with $I \subseteq \mathfrak{m}$.

Consider $x/1 \in M_{\mathfrak{m}}$. Since $M_{\mathfrak{m}} = 0$, $x/1 = 0$, so there is $t \in R \setminus \mathfrak{m}$ with $tx = 0$. Then $t \in I \subseteq \mathfrak{m}$ – contradicting $t \notin \mathfrak{m}$. □

Exercise 1.56. Let $\varphi \in \operatorname{Hom}(M, N)$. TFAE:

(1) φ is injective (resp. surjective);

(2) $\varphi_P : M_P \to N_P$ is injective (resp. surjective) for all $P \in \operatorname{Spec}(R)$;

(3) $\varphi_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ is injective (resp. surjective) for all $\mathfrak{m} \in \operatorname{Max}(R)$.

Proposition 1.57. Let $R \hookrightarrow S$ be rings with S integral over R.

(1) If J is an ideal of S, then $R/(J \cap R) \hookrightarrow S/J$ is integral.

(2) If T is a multiplicative subset of R, then $T^{-1}R \hookrightarrow T^{-1}S$ is integral.

Proof. (1) is straightforward (reduce the monic integral equation modulo J).

(2) Let $x/t \in T^{-1}S$ ($x \in S$, $t \in T$). Since x is integral over R,

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0 \qquad \text{for some } a_i \in R.$$

Dividing by t^n,

$$\left(\frac{x}{t}\right)^n + \frac{a_1}{t}\left(\frac{x}{t}\right)^{n-1} + \cdots + \frac{a_n}{t^n} = 0,$$

so x/t is integral over $T^{-1}R$. □

Proposition 1.58. Let $R \hookrightarrow S$ be an integral extension, with R, S both integral domains. Then R is a field if and only if S is a field.

Proof. ($\Rightarrow$) Let R be a field and $y \in S \setminus \{0\}$. Since y is integral over R, take a monic relation of minimal degree,

$$y^n + a_1 y^{n-1} + \cdots + a_n = 0, \qquad a_i \in R.$$

If $a_n = 0$, then $y(y^{n-1} + a_1 y^{n-2} + \cdots + a_{n-1}) = 0$; since S is a domain and $y \neq 0$, this gives a relation of degree $n-1$, contradicting minimality. So $a_n \neq 0$, and as R is a field, a_n is a unit. Then

$$y\left(y^{n-1} + \cdots + a_{n-1}\right) = -a_n,$$

and since $-a_n$ is a unit, y is a unit.

($\Leftarrow$) Let S be a field. Take any $x \in R \setminus \{0\}$. Then $x^{-1} \in S$ is integral over R:

$$x^{-m} + a_1 x^{-(m-1)} + \cdots + a_m = 0, \qquad a_i \in R,$$

so

$$x^{-1} = -\left(a_1 + a_2 x + \cdots + a_m x^{m-1}\right) \in R.$$

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CHAPTER 2

PROBLEM DISCUSSIONS

2.1 Assignment 0

Convention. By a ring, we mean a commutative ring with 1. Also, we exclude the zero ring.

1. Let I be an ideal of R and $\operatorname{rad}(I) := \{a \in R \mid a^n \in I \text{ for some } n \geq 1\}$. Prove the following:

   (a) $\operatorname{rad}(\operatorname{rad}(I)) = \operatorname{rad}(I)$

   Solution. The containment $\operatorname{rad}(I) \subseteq \operatorname{rad}(\operatorname{rad}(I))$ is immediate, since $I \subseteq \operatorname{rad}(I)$ and $\operatorname{rad}(-)$ is monotone.

   Conversely, let $x \in \operatorname{rad}(\operatorname{rad}(I))$. Then $x^n \in \operatorname{rad}(I)$ for some n, so $(x^n)^m = x^{mn} \in I$ for some m. Hence $x \in \operatorname{rad}(I)$. $\square$

   (b) $\operatorname{rad}(IJ) = \operatorname{rad}(I \cap J) = \operatorname{rad}(I) \cap \operatorname{rad}(J)$

   Solution. We show $\operatorname{rad}(I) \cap \operatorname{rad}(J) \subseteq \operatorname{rad}(IJ) \subseteq \operatorname{rad}(I \cap J) \subseteq \operatorname{rad}(I) \cap \operatorname{rad}(J)$, giving all three sets equal.

   - $\operatorname{rad}(I) \cap \operatorname{rad}(J) \subseteq \operatorname{rad}(IJ)$: if $x \in \operatorname{rad}(I) \cap \operatorname{rad}(J)$, then $x^n \in I$ and $x^m \in J$ for some n, m, so $x^{n+m} = x^n x^m \in IJ$, i.e. $x \in \operatorname{rad}(IJ)$.
   - $\operatorname{rad}(IJ) \subseteq \operatorname{rad}(I \cap J)$: since $IJ \subseteq I \cap J$, this follows from monotonicity of rad.
   - $\operatorname{rad}(I \cap J) \subseteq \operatorname{rad}(I) \cap \operatorname{rad}(J)$: since $I \cap J \subseteq I$ and $I \cap J \subseteq J$, monotonicity gives $\operatorname{rad}(I \cap J) \subseteq \operatorname{rad}(I)$ and $\operatorname{rad}(I \cap J) \subseteq \operatorname{rad}(J)$.

   $\square$

   (c) $\operatorname{rad}(I) = R$ if and only if $I = R$

   Solution. ($\Rightarrow$) If $\operatorname{rad}(I) = R$, then for every $a \in R$ there is $n \in \mathbb{N}$ with $a^n \in I$. Taking $a = 1$: $1^n = 1 \in I$, so $I = R$.

   ($\Leftarrow$) Trivial: if $I = R$ then $\operatorname{rad}(I) = R$ as well. $\square$

   (d) $\operatorname{rad}(I + J) = \operatorname{rad}(\operatorname{rad}(I) + \operatorname{rad}(J))$

Solution. ($\subseteq$) Let $x \in \mathrm{rad}(I+J)$. Then $x^k = u + v$ for some $k \in \mathbb{N}$, $u \in I$, $v \in J$. Since $u \in I \subseteq \mathrm{rad}(I)$ and $v \in J \subseteq \mathrm{rad}(J)$, $x^k \in \mathrm{rad}(I) + \mathrm{rad}(J)$, so $x \in \mathrm{rad}(\mathrm{rad}(I) + \mathrm{rad}(J))$.

($\supseteq$) Since $I \subseteq I + J$, monotonicity gives $\mathrm{rad}(I) \subseteq \mathrm{rad}(I+J)$; similarly $\mathrm{rad}(J) \subseteq \mathrm{rad}(I+J)$. As $\mathrm{rad}(I+J)$ is an ideal, $\mathrm{rad}(I) + \mathrm{rad}(J) \subseteq \mathrm{rad}(I+J)$. Applying $\mathrm{rad}(-)$ and using part (a),

$$\mathrm{rad}(\mathrm{rad}(I) + \mathrm{rad}(J)) \subseteq \mathrm{rad}(\mathrm{rad}(I+J)) = \mathrm{rad}(I+J).$$

□

(e) If $P \in \mathrm{Spec}(R)$, then $\mathrm{rad}(P^n) = P$ for all $n > 0$

Solution. First, $\mathrm{rad}(P) = P$: the containment $P \subseteq \mathrm{rad}(P)$ always holds, and $\mathrm{rad}(P) \subseteq P$ follows since P is prime (if $x^n \in P$, primality forces $x \in P$).

Now induct on n. The base case $n = 1$ is $\mathrm{rad}(P) = P$, shown above. For the inductive step, using part (b) with $I = P^{n-1}$, $J = P$:

$$\mathrm{rad}(P^n) = \mathrm{rad}(P^{n-1} \cdot P) = \mathrm{rad}(P^{n-1}) \cap \mathrm{rad}(P) = P \cap P = P,$$

using the induction hypothesis $\mathrm{rad}(P^{n-1}) = P$. □

2. Let $R[X]$ be the polynomial ring over a ring R. Prove that the nilradical of $R[X]$ is equal to the Jacobson radical of $R[X]$.

Solution. Since every maximal ideal is prime, intersecting over the (larger) family of prime ideals gives a smaller or equal intersection than intersecting over maximal ideals alone; hence, for any ring, $\mathcal{N}(R[X]) \subseteq J(R[X])$.

For the reverse containment, we use the identity $\mathcal{N}(R[X]) = (\mathcal{N}(R))[X]$ (proof deferred). Take $f(X) \in J(R[X])$. By the standard characterization of the Jacobson radical, $1 - g(X)f(X)$ is a unit for every $g(X) \in R[X]$; taking $g(X) = -X$,

$$1 + Xf(X) \text{ is a unit in } R[X].$$

By the Lecture 1 proposition on units of $R[X]$, since the constant term of $1 + Xf(X)$ is 1 (a unit), all remaining coefficients of $1 + Xf(X)$ – i.e. all coefficients of $f(X)$ – are nilpotent. Hence $f(X) \in (\mathcal{N}(R))[X] = \mathcal{N}(R[X])$.

Therefore $J(R[X]) \subseteq \mathcal{N}(R[X])$, and combined with the reverse containment above, $\mathcal{N}(R[X]) = J(R[X])$. □

3. Let $R[X]$ be the polynomial ring over a ring R. Let $J \subset R$ be an ideal. Define

$$J[X] := \{a_0 + a_1 X + \cdots + a_n X^n \mid a_i \in J \;\forall\, i\}.$$

Prove that $J[X]$ is an ideal of $R[X]$. Give an example of an ideal of $R[X]$ which is not of this form. Prove that the rings $R[X]/J[X]$ and $(R/J)[X]$ are isomorphic.

Solution. That $J[X]$ is an ideal of $R[X]$ is a direct verification (closed under addition and absorbs multiplication by $R[X]$, coefficientwise, since J is an ideal of R).

Example of an ideal not of this form. Let

$$I = \{p(X) \in R[X] \mid \exists\, q(X) \in R[X] \text{ such that } p(X) = X \cdot q(X)\} = \langle X \rangle,$$

the ideal of polynomials with zero constant term. This is not of the form $J[X]$ for any proper ideal $J \subsetneq R$: an element of $J[X]$ has *every* coefficient in J, whereas I places no constraint on coefficients beyond the constant term.

Isomorphism. Define $\varphi : R[X]/J[X] \to (R/J)[X]$ by

$$\varphi\big(r(X) + J[X]\big) = \sum_{i=0}^{\deg r} (r_i + J)\, X^i,$$

i.e. reduce each coefficient of $r(X)$ modulo J. This is induced by the coefficientwise reduction map $R[X] \to (R/J)[X]$, which is a surjective ring morphism with kernel exactly $J[X]$; by the First Isomorphism Theorem, φ is a well-defined ring isomorphism. □

4. A commutative ring is called a *local ring* if it has only one maximal ideal. Let R be a local ring with unique maximal ideal $\mathfrak{m}$. Prove that any element $a \in R \setminus \mathfrak{m}$ is a unit in R. Conversely, let R be a ring such that the set of non-units forms an ideal M. Prove that R is local with maximal ideal M.

Solution. **(⇒)** FTSOC, suppose $a \in R \setminus \mathfrak{m}$ is not a unit. Then $\langle a \rangle \neq R$, i.e. $1 \notin \langle a \rangle$. Let

$$A = \{I \subseteq R \text{ an ideal} \mid a \in I \text{ but } 1 \notin I\}.$$

Since $\langle a \rangle \in A$, $A \neq \emptyset$. By Zorn's Lemma, let M be a maximal element of A (ordered by inclusion). One checks M is a maximal ideal of R. Since R is local, $M = \mathfrak{m}$, and since $a \in M$, we get $a \in \mathfrak{m}$ – contradicting $a \notin \mathfrak{m}$. Hence a must be a unit.

(Converse) Suppose $M = \{x \in R \mid x \text{ is not a unit}\}$ is an ideal. Every proper ideal I of R consists entirely of non-units (else I would contain a unit, forcing $I = R$), so $I \subseteq M$. In particular M itself is a proper ideal (as $1 \notin M$), and since every proper ideal is contained in M, M is the unique maximal ideal of R. Hence R is local with maximal ideal M. □

5. An element $e \in R$ is called an *idempotent* if $e^2 = e$. Let R be a local ring. Prove that the only idempotents of R are 0 and 1.

Solution. **Lemma.** If R is a local ring, then for every $a \in R$, either a or $1 - a$ is a unit.

Proof. Suppose neither a nor $1 - a$ is a unit. Then $a \in \mathfrak{m}$ and $1 - a \in \mathfrak{m}$ (the unique maximal ideal), so $1 = a + (1 - a) \in \mathfrak{m}$, forcing $\mathfrak{m} = R$ – a contradiction.

Now let $e \in R$ be idempotent: $e^2 = e$, i.e. $e(e-1) = 0$. By the lemma, at least one of $e, 1-e$ is a unit; multiplying $e(e-1) = 0$ by its inverse forces the other factor to vanish. Hence $e = 0$ or $e = 1$. $\square$

6. Let $R = \mathbb{Z}[i]$. Prove that the ideal $P = \langle 1+i \rangle$ is a prime ideal of R. Prove that $\langle 2 \rangle = P^2$ in R.

7. Prove that the ring $\mathbb{Z}[i]/\langle 2+i \rangle$ is isomorphic to the ring $\mathbb{Z}/5\mathbb{Z}$.

8. Let R be a ring and $a, b \in R$ be two *comaximal* elements, i.e., there exist $x, y \in R$ such that $xa + yb = 1$. Prove that for any $m, n \in \mathbb{N}$, one has a^m and b^n are also comaximal (i.e., there exist $\lambda, \mu \in R$ such that $\lambda a^m + \mu b^n = 1$).

9. Let R be an integral domain. Let $a, b \in R$ be such that their gcd exists. If d, d' are both gcds of a, b, prove that d and d' are associates.

10. Let R be an integral domain such that $\gcd(a, b)$ exists for any $a, b \in R \setminus \{0\}$. Then prove that $\operatorname{lcm}(a, b)$ exists for any $a, b \in R \setminus \{0\}$. (Hint: First prove that $\gcd(ma, mb) = m\gcd(a, b)$.)

11. Let R be an integral domain such that $\operatorname{lcm}(a, b)$ exists for any $a, b \in R \setminus \{0\}$. Then prove that $\gcd(a, b)$ exists for any $a, b \in R \setminus \{0\}$.

12. Let R be an integral domain such that $\gcd(a, b)$ exists for any $a, b \in R \setminus \{0\}$. If $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$, then prove that $\gcd(a, bc) = 1$.

13. Let $R = \mathbb{Z}[\sqrt{-5}]$. Verify the following:

    (a) There are two distinct (i.e. non-associate) factorizations of 6 into irreducibles in R:

    $$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}).$$

    (b) The ideals $\langle 2, 1 \pm \sqrt{-5} \rangle$, $\langle 3, 1 \pm \sqrt{-5} \rangle$ are all prime ideals.

    (c) The following factorization of ideals into prime ideals holds:

    - $\langle 2 \rangle = \langle 2, 1 + \sqrt{-5} \rangle^2$,
    - $\langle 3 \rangle = \langle 3, 1 + \sqrt{-5} \rangle \langle 3, 1 - \sqrt{-5} \rangle$,
    - $\langle 1 + \sqrt{-5} \rangle = \langle 2, 1 + \sqrt{-5} \rangle \langle 3, 1 + \sqrt{-5} \rangle$,
    - $\langle 1 - \sqrt{-5} \rangle = \langle 2, 1 + \sqrt{-5} \rangle \langle 3, 1 - \sqrt{-5} \rangle$.

    (d) Uniqueness of factorization is restored in (a) at the level of ideals.

    (e) The ideal $\langle 2, 1 + \sqrt{-5} \rangle$ is not a principal ideal.

14. Show that $\langle 2, 1 + \sqrt{-17} \rangle$ is not a principal ideal in $\mathbb{Z}[\sqrt{-17}]$. Is this a prime ideal?

15. Let R be an ID. Assume that:

    (a) any two $a, b \in R \setminus \{0\}$ have gcd, say, d, and there are $x, y \in R$ such that $d = ax + by$;

(b) if $a_1, a_2, \ldots$ are nonzero elements of R such that a_{i+1} divides a_i for $i = 1, 2, \ldots$, then there is $N > 0$ such that $a_n = u_n a_N$ for some unit $u_n \in R$ for all $n \geq N$.

Show that R is a PID.

16. Prove that $R[X]/\langle X^3 + X \rangle$ is a product of fields.

17. Let $m, n \in \mathbb{Z}$ be two nonzero elements and $d = \gcd(m, n)$ in $\mathbb{Z}$. Prove that d is also a gcd of m, n in $\mathbb{Z}[i]$.

18. Let $m, n \in \mathbb{Z}$ be two nonzero elements and let m divide n in $\mathbb{Z}[i]$. Does m also divide n in $\mathbb{Z}$?

19. Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of R-modules. Let M', M'' be both finitely generated. Show that M is finitely generated.

20. From Dummit–Foote: Exercise 10.1: 5, 8, 9, 10, 11, 15.

21. From Dummit–Foote: Exercise 10.2: 3, 4, 5, 6, 7, 9, 10, 11, 12, 13.

22. An R-module is called *simple* if it is not the zero module and it has no proper submodule.

    (a) Prove that an R-module M is simple if and only if it is isomorphic (as an R-module) to $R/\mathfrak{m}$, where $\mathfrak{m}$ is a maximal ideal of R. Show also that M is generated by any non-zero element.

    (b) Let M, N be simple R-modules. Prove that a module morphism $\phi : M \to N$ is either the zero morphism or an isomorphism. Conclude that if M is simple, $\mathrm{Hom}_R(M, M)$ is a division ring.

23. Let M be an additive abelian group and $\mathrm{End}(M)$ be the set of all group morphisms from M to M.

    (a) Show that $\mathrm{End}(M)$ is a ring with identity (where addition is defined pointwise, and multiplication is composition of maps).

    (b) Show that M is an $\mathrm{End}(M)$-module under the action:

    $$(f, m) \mapsto fm := f(m).$$

    (c) Let R be a ring and $\phi : R \to \mathrm{End}(M)$ be a ring morphism. Show that M is an R-module under the action:

    $$(r, m) \mapsto rm := \phi(r)(m).$$

24. (Converse of 4(c)): Let R be a ring and M be an R-module. Show that there is a ring morphism $\phi : R \to \mathrm{End}(M)$.

25. Let M be an R-module and A, B, C be its submodules such that

$$A \subseteq B, \qquad A + C = B + C, \qquad A \cap C = B \cap C.$$

Prove that $A = B$.

26. Let $f : M \to N$ be an R-module morphism. Let A be a submodule of M. Prove that $f^{-1}(f(A)) = A + \mathrm{Ker}(f)$. Now let B be a submodule of N. Prove that $f(f^{-1}(B)) = B \cap \mathrm{Im}(f)$. Also, show that
$$f(A \cap f^{-1}(B)) = f(A) \cap B.$$

27. Let R be a commutative ring. Show that a map $f : R \times R \to R$ is an R-module morphism if and only if $f(x, y) = \alpha x + \beta y$ for some $\alpha, \beta \in R$.

28. Let $f : M \to N$ be an R-module morphism. Show that f can be expressed as the composition of three R-module morphisms (for appropriate R-modules): $f = g \circ h \circ k$, where g is surjective, h is an isomorphism, and k is injective.

29. Let R be an integral domain and $a \in R$ be non-zero. Let n be any positive integer. Prove that Ra^n / Ra^{n+1} is isomorphic to R/Ra (as R-modules).

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